MATEO LÓPEZ

COMPUTATIONAL DESIGNER · RHINO / GRASSHOPPER · DYNAMO · REVIT API · C# / PYTHON

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English (TOEFL B2) · Spanish (native)

Portfolio · LinkedIn · GitHub / matbimdev

PROFILE

Civil engineer and computational designer who turns structural, facade and MEP engineering rules into parametric models and reusable tools. Five years in the built environment: bridge analysis and design in MIDAS Civil, BIM coordination on $88M of high-rise residential construction, and now automation lead for a multi-discipline Scan-to-BIM team delivering U.S. projects. Builds in Rhino, Grasshopper, Rhino.Inside.Revit, Dynamo, Python and C# on the Revit API, using Grasshopper as the bridge between parametric geometry and native Revit elements, with 25+ tools and 50+ Dynamo scripts in daily production and 40+ colleagues trained to use them.

KEY ACHIEVEMENTS

- **MEP routing from CAD to Revit:** fire-protection networks with automatic elbow, tee, cross and tap fittings, from 8–12 hours to under 2 minutes per floor.
- **Parametric structural workflows in Grasshopper, Rhino.Inside.Revit and Dynamo** that turn live models and DWG drawings into formwork, rebar, column sections, footings and quantity takeoffs.
- **Staged erection of a tied-arch bridge** modeled in MIDAS Civil to AASHTO LRFD, up to a 29.38 m cantilever, including temporary towers and their foundations.
- **40+ engineers and modelers trained** across MEP, FP, architecture and structure, backed by documentation for 25+ pyRevit tools, 50+ Dynamo scripts and 20+ video tutorials.

TECHNICAL SKILLS

Computational design: Rhino · Grasshopper (Python 3 script components, Rhino.Inside.Revit, DirectShape, eleFront, LunchBox, Galapagos) · Dynamo (2020–2026) & Dynamo Player · pyRevit

Programming & APIs: Python (IronPython, CPython) · C# / .NET (WPF) · Revit API · Rhino.Inside.Revit API

Structural analysis: MIDAS Civil · MIDAS Gen · CSI Bridge · SAP2000 · ETABS · AASHTO LRFD

BIM & coordination: Revit Structure, MEP & Architecture (advanced) · Tekla Structures · Civil 3D · Navisworks · AutoCAD · ACC / BIM 360 · IFC · clash detection · Scan-to-BIM

Data: Excel ↔ Revit pipelines · Power BI · JSON · Git & GitHub

PROFESSIONAL EXPERIENCE

BIM Specialist, Automation | Hexagon Multivista, formerly Voyansi (USA, remote) **Jul 2025 – Present**

Automation lead for a Scan-to-BIM production team modeling MEP, fire protection, architecture and structure for U.S. commercial, industrial, healthcare and sports projects (30,000 to 1,000,000+ ft² each).

- **Developed 25+ pyRevit tools and 50+ Dynamo scripts** used daily across all disciplines and QC, cutting manual modeling and data work by 40–60% per task.
- **Automated MEP routing from CAD** (pipes, conduits, sprinkler connections) and solved an undocumented Revit API limitation to tap pipes across different MEP systems with 100% reliability.
- **Built a parametric geometry pipeline** with Boolean solid operations, face projection and worksharing-aware transactions that generates ceilings, floors and finishes locked to their host walls.
- **Turned written BIM standards into automated model scans:** MEP clearance under ceilings and level assignment within ±80 mm.
- **Trained 40+ team members** and wrote the documentation and 20+ video tutorials for every tool, deployed through Dynamo Player so non-developers run them unassisted.

BIM Technician, Scan-to-BIM | Hexagon Multivista (USA, remote) **Nov 2024 – Jun 2025**

- **Converted laser-scan point clouds** into multi-discipline models at LOD 200–400 on industrial buildings of 30,000–60,000 m².
- **Promoted to automation within 6 months** after building a tool that measures wall thickness from the point cloud and creates the matching wall type in three clicks.

Freelance Computational Design & Automation | International clients **2025 – Present**

- **Structure:** parametric sports pavilion generator in Dynamo where 26 inputs drive levels, grids, slabs, walls, columns and a curved roof of adaptive beams; automated column section documentation within ±40 mm (Saudi Arabia); parametric retaining wall families.
- **Facade:** catalog-driven panel and gutter placement around openings (Netherlands); parametric louver study comparing four solar-control configurations (UAE); glazing ratio analysis (France).
- **MEP:** rule-based socket, sanitary and ventilation layouts for residential buildings with 80+ obstacle checks per room (Germany).
- **Data:** toposolid elevation heat map (UK) and material quantity takeoff to Excel (France).

Founder & Lead Developer | AUBIMAT (aubimat.com) **2025 – Present**

- **C#/.NET Revit add-in with 13 commands** across architecture, structure, MEP, annotation, QC and views; one codebase for Revit 2022–2027 (about 21,000 lines of C#) with WPF interfaces and an installer.
- **CAD-to-BIM generators:** structural columns (rectangular, circular, L-shaped) and walls (L, T and cross junctions) from DWG layers with automatic type creation, fire-protection collector piping with elbows, tees, crosses and taps, sprinklers connected to the piping, families from CAD blocks, plus a clash browser and a level fixer for model QC.
- **Wall-envelope geometry engine** for floors, ceilings and finish walls: Boolean solid union and enclosed-region detection, with an interactive plan preview that flags open wall ends and suggests trim or extend fixes before creating elements locked to their host walls.

BIM Coordinator | Diseño Meta S.A.C. (Lima, Peru) **Sep 2022 – Oct 2024**

- **Coordinated design and construction of $88M of residential work:** the 12-tower Bicentenario complex (1,350,000 ft^2) and the Central Park (22 floors) and Inspiria (20 floors) high-rises.
- **Led ICE sessions with architects and structural and MEP engineers** for clash detection and design resolution in Navisworks and BIM 360; managed federated models, RFIs and LOD 300 tracking.
- **Authored BIM Execution Plans and led a team of BIM modelers;** audited construction plans against contract documents and produced quantities, layout models and As-Built documentation.

BIM Modeler | Adinnova S.A.C. (Lima, Peru) **Mar 2022 – Aug 2022**

- **Modeled two hospitals, a multifamily building and a school** at LOD 300 across structure, architecture and MEP.

Structural Engineering Intern, Bridges | Gestruc S.A.C. (Lima, Peru) **Aug 2021 – Nov 2021**

- **Modeled the staged erection of the Santa Rosa tied-arch bridge** in MIDAS Civil to AASHTO LRFD: launch stages up to a 29.38 m cantilever, temporary towers, stay cables and foundations.
- **Designed falsework elements for a 120 m-span steel arch bridge** under wind and seismic loads; produced drawings and quantity takeoffs.

SELECTED COMPUTATIONAL PROJECTS

Full case studies with code and results: matbimdev.github.io/Portfolio

- **Formwork and rebar from live structural models:** Grasshopper and Rhino.Inside.Revit definitions that read Revit structural elements to generate formwork surfaces and contact areas (eleFront) and rebar, with structural takeoffs computed by Python 3 components on the Revit API.
- **Parametric acoustic ceiling, design to production:** double-curvature surface rationalized into panels in Grasshopper and pushed to Revit adaptive families with IFC property mapping via Rhino.Inside.Revit.
- **Footing and column generator:** reads structural DWG drawings and builds parametric 3D footings and columns with rebar and quantity reports, using Python 3 components in Grasshopper.
- **Point cloud to BIM:** semi-automatic extraction of columns, walls and pipes from scans into Revit with Dynamo.

EDUCATION & CERTIFICATIONS

Master in International BIM Management | EADIC, Spain **2026**

B.Sc. Civil Engineering | Universidad Nacional de Ingeniería, Peru **2016 – 2021**

- **Grasshopper in Revit Training (2026):** element creation, parameters, views and sheets, annotations, DirectShape and adaptive components · Dynamo Specialist: Dynamo, Python and Revit API (2025)
- **AECODE Specialization in Process Automation and AI Agents (2026, 100 h, endorsed by CIP and Autodesk ATC):** Generative AI for Workflow Optimization · Process Automation with Make, n8n and Zapier · AI Assistants and Agents Development (MCP, tool calling)
- **Autodesk ATC (2025):** BIM Coordinator with Revit, Navisworks & Power BI · BIM 4D Planning with Navisworks (120 h each) · buildingSMART BIM Fundamentals · Lean Construction (150 h) · NEC Contracts (120 h)